WILP, BITS Pilani

MACHINE LEARNING · COMPANION READER

Naïve Bayes & Optimal Classifiers

MAP Estimation, Bayes Optimal Classifier, Gibbs Classifier, and Naïve Bayes Classifier

Session 11 · Read alongside the lecture slides · Every slide example worked out in full

Prepared for the Work Integrated Learning Programmes (WILP), BITS Pilani.

1 MAP Estimation Worked Examples

MAP estimation maximizes the product of likelihood and prior probability density.

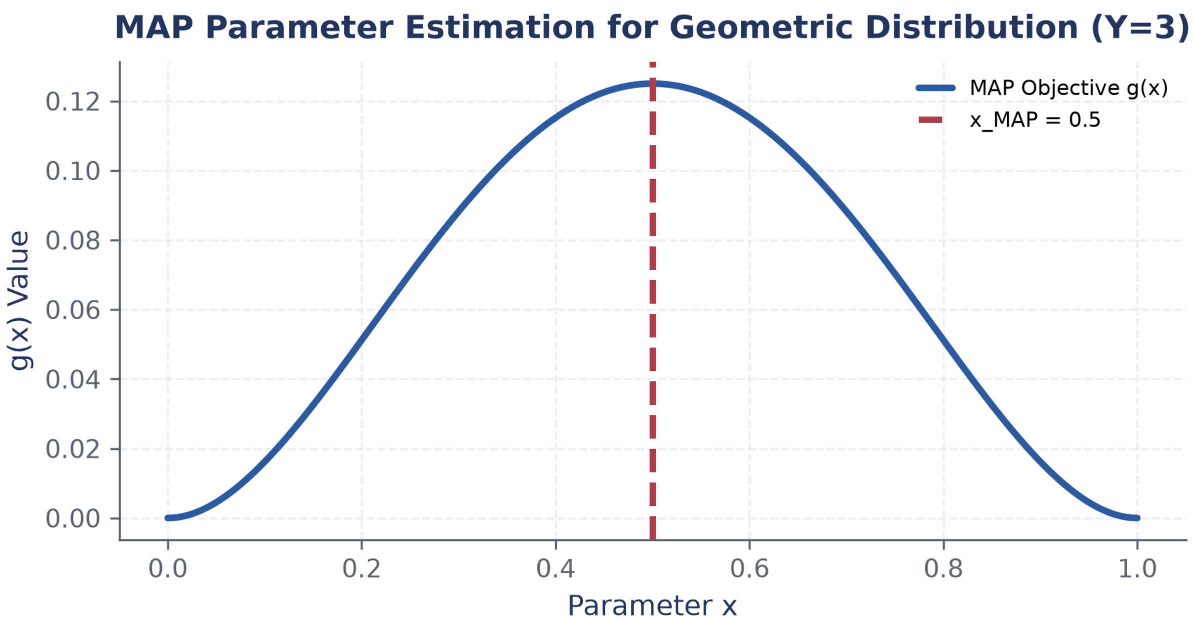

Figure 1: Unnormalized MAP objective function for the geometric distribution.

✎ Worked example: Geometric Distribution MAP Parameter Estimation

Geometric likelihood $P(Y = y \mid X = x) = x(1-x)^{y-1}$ with prior $P(X = x) = 2x$. Given $Y = 3$:

Step 1. $g(x) = 2x^2(1-x)^2 \implies \frac{d \ln g(x)}{dx} = \frac{2}{x} - \frac{2}{1-x} = 0 \implies x_{\text{MAP}} = 0.5.$

2 Bayes Optimal Classifier & Gibbs Classifier

The Bayes Optimal Classifier achieves minimum error rate.

3 Naïve Bayes Classifier

Naïve Bayes assumes conditional independence among features.

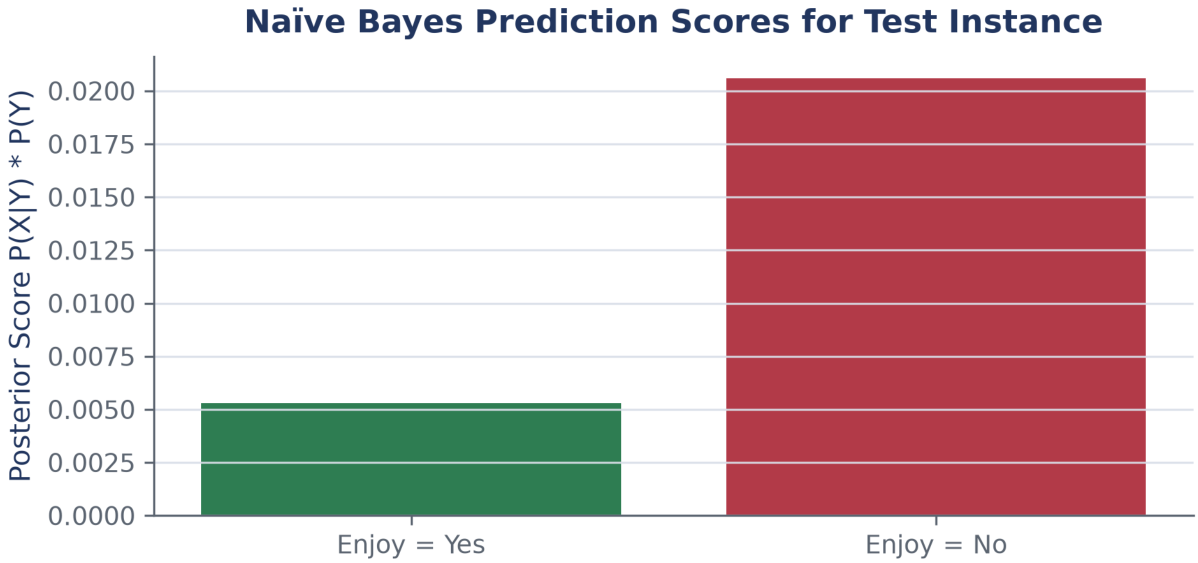

Figure 2: Naïve Bayes posterior prediction scores.

4 Laplace Smoothing (m-estimate)

Laplace smoothing prevents zero probabilities.

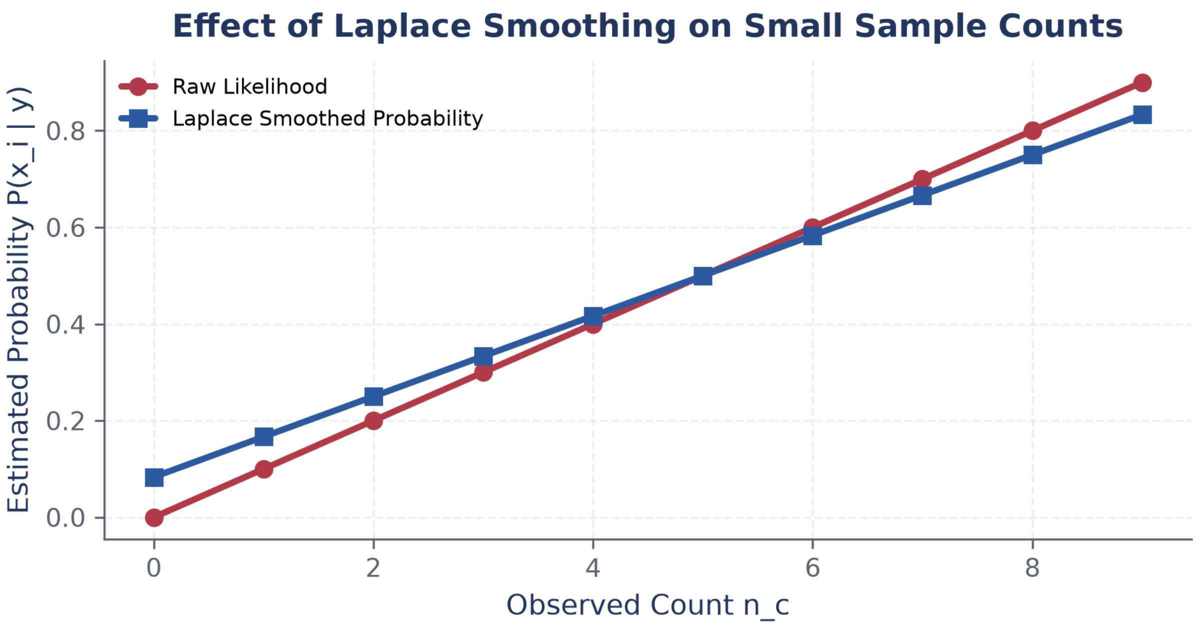

Figure 3: Effect of Laplace smoothing on estimated feature probabilities.